

Design Document: Introduction to Game Programming

Working-Title: Thrust-Clone

Arsenii Parhom; 2606295
Irmak Damla Özdemir; 2606395
Max Tremel; 2606396

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[1] This features belong to Irmak Damla Özdemir

Feature 2: Combat & Damage System

1. What the feature does and what it brings to the game

This feature handles how health and damage work for both the player and the turrets, and shows this information to the player through the HUD. Damage is triggered whenever a projectile fired by a turret collides with the rocket, and the resulting health changes are reflected visually so the player always knows how much danger they are in. For the player, health and damage are shown as a bar in the HUD, as illustrated in Figure 1. For turrets, damage is planned to be shown directly above the turret itself, as a red horizontal bar. This gives combat encounters real consequences and helps the player make informed decisions, such as when to retreat or when it is safe to push forward.



Figure 1: Health Bar Examples

2. Brief Architecture

The core part for the feature is a health component that can be attached to any entity that should be able to take damage. It keeps track of current health, max health and has function to apply damage like `ApplyDamage(...)`. When the damage function is called, it lowers the entity's current health and makes sure it stays in a consistent interval.

After the health changes, the component sends out signal, so anything in the system that cares about health changes can just listen for this signal and react to it. If health reaches zero, a separate signal like `Died` is sent. If it is for the rocket, then game is over, if it is for a turret, then the turret is removed.

The HUD is built on top of this in a similar way. It doesn't check the player's health every frame. It just subscribes to the signal from the player's health component and updates the health bar whenever that signal fires. This aims to keep the HUD completely separate from combat logic: it only reads and displays data, it never changes health values itself.

3. Dependencies and Risks

One risk is that a single hit could accidentally count as multiple hits if a collision triggers more than once in the same frame. Therefore, a short period of invincibility after getting hit might be needed.

Another point that will likely require adjustment is balancing the actual damage and health values across the rocket and turrets, which is expected to be an iterative process based on playtesting.

4. Acceptance Criteria

- Every that can take damage has a current and max health value.

- Taking damage reduces health by the correct amount and health never goes below zero or above max.
- When health reaches zero by the player, then the game is over for the player.
- When health reaches zero by a turret, then the turret is removed / vanished.
- The HUD displays the player's health correctly and updates right away when it changes.
- Turret's damage / health is planned to shown on the top itself as a red horizontal bar. Yet, can be changed depending on the adjustments.